

From 3D Pose to Prose: Biomechanics-Grounded Vision–Language Coaching

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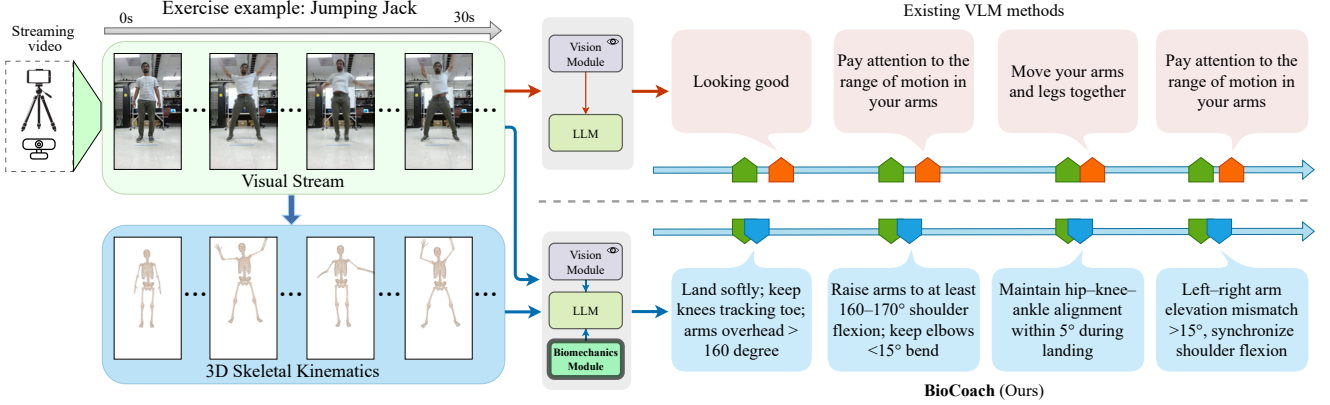


Figure 1. Comparison with existing methods. Top: prior pixel-only VLM methods provide generic, loosely timed comments. Bottom: **BioCoach** fuses visual features with 3D skeletal kinematics and a biomechanics module to produce phase-aligned, anatomy-specific, quantitative cues (e.g., shoulder flexion 160° – 170°), yielding more precise and biomechanics-grounded feedback along the same timeline.

Abstract

We present *BioCoach*, a biomechanics-grounded vision–language framework for fitness coaching from streaming video. *BioCoach* fuses visual appearance and 3D skeletal kinematics, through a novel three-stage pipeline: an exercise-specific degree-of-freedom selector that focuses analysis on salient joints; a structured biomechanical context that pairs individualized morphometrics with cycle and constraint analysis; and a vision–biomechanics conditioned feedback module that applies cross-attention to generate precise, actionable text. Using parameter-efficient training that freezes the vision and language backbones, *BioCoach* yields transparent, personalized reasoning rather than pattern matching. To enable learning and fair evaluation, we augment *QEVD-fit-coach* with biomechanics-oriented feedback to create *QEVD-bio-fit-coach*, and we introduce a biomechanics-aware LLM judge metric. *BioCoach* delivers clear gains on *QEVD-bio-fit-coach* across lexical and judgment metrics while maintaining temporal triggering; on the original *QEVD-fit-coach*, it improves text quality and correctness with near-parity timing, demonstrating that explicit kinematics and constraints are key to accurate, phase-aware coaching. [Project](#)

1. Introduction

Live fitness coaching through streaming video is increasingly important, with a broad impact on at-home workouts [44], injury prevention [15, 31], and data-driven rehabilitation [1, 5, 38]. Traditionally, fitness coaching has relied on in-person trainers who provide immediate feedback to correct form deviations and reduce risks; however, expert coaching is expensive and often inaccessible. Advances in computer vision now enable automated form assessment from video [13, 36], yet existing approaches face three critical challenges that are also the field’s key requirements: (1) Timing and interaction, detecting coachable moments and delivering feedback at the right time rather than post hoc; (2) Biomechanical grounding, reasoning over 3D poses, joint angles, ranges of motion, and exercise phases instead of high-level appearance; and (3) Personalization and explainability, producing traceable, rule-level evidence.

Recent vision-language models have revitalized multi-modal understanding [4, 42, 48] and can generate fluent, instruction-like feedback [8, 27]; streaming VLMs such as Stream-VLM [32] apply these capabilities to live fitness coaching. However, they remain largely prompt-driven and struggle to autonomously surface coachable moments; they lack explicit morphometric context for personalization; and they do not integrate symbolic biomechanical constraints, which can yield generic or mistimed advice (see Fig. 1).

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To address these limitations, we propose **BioCoach**, a *biomechanics-grounded vision-language* framework for streaming fitness coaching that bridges biomechanical analysis and multimodal understanding. Our core insight is to construct *explicit, interpretable intermediate representations* that expose kinematic properties to the language model, enabling transparent reasoning about exercise form while preserving end-to-end trainability. Rather than treating visual appearance and 3D pose as disjoint streams or relying solely on pattern learning, BioCoach organizes them into a structured pipeline that grounds feedback in biomechanical principles and individual morphology (see Fig. 1).

Specifically, BioCoach extracts two complementary signals from streaming video: a visual stream that captures appearance and context, and a 3D kinematic stream that captures skeletal pose and body shape. The system then proceeds through three coordinated steps, each introducing a novel design. *First*, an exercise-specific joint attention mechanism prioritizes the degrees of freedom relevant to the current exercise, focusing analysis on anatomically salient regions instead of treating all joints uniformly. *Second*, a structured biomechanical context pairs morphometric information for personalization with cycle analysis against curated references to yield explicit, rule-based form cues, which replace implicit visual heuristics with verifiable evidence. *Third*, a vision-biomechanics conditioning mechanism fuses visual evidence with this context to produce coaching that is temporally targeted, interpretable through intermediate representations, and grounded in biomechanical principles rather than relying solely on pattern matching. Moreover, to support learning and evaluation, we create *QEVD-bio-fit-coach* by re-annotating the QEVD-fit-coach dataset [32] with fine-grained biomechanical annotations.

In summary, the contributions of this work include:

- ◊ We propose **BioCoach**, a new biomechanics-grounded vision-language framework for interpretable, personalized coaching from streaming video.
- ◊ We devise a three-stage pipeline unifying exercise-specific joint attention, biomechanical context for personalized rule-level cues, and vision-biomechanics conditioned feedback for targeted, auditable coaching.
- ◊ We create *QEVD-bio-fit-coach* by re-annotating QEVD-fit-coach with fine-grained biomechanical labels.
- ◊ Extensive experiments show that **BioCoach** achieves superior text quality, timing accuracy, LLM-graded correctness, and biomechanics-grounded action quality scores.

2. Related Work

Interactive Fitness Coaching and Vision-Language Models. Interactive fitness coaching from video has emerged as a key research direction with benchmarks like QEVD [32] that pair long-range videos with times-

tamped coaching feedback. Recent vision-language models—including Flamingo [2], InstructBLIP [8], LLaVA variants [23, 51], and LLaMA-VID [21]—demonstrate strong video understanding via gated cross-attention and token-level fusion. Stream-VLM [32], a strong QEVD baseline, introduced asynchronous interaction with action tokens to enable unprompted feedback. However, these models operate primarily on pixel-level features without explicit biomechanical constraints, leading to feedback that can be generic or poorly timed. BioCoach addresses this by extracting and integrating 3D skeletal kinematics as a structured modality, enabling biomechanics-grounded coaching.

Pose, Shape Estimation, and Biomechanical Analysis. Advances in 3D human pose and mesh recovery span classic HMR [17], PARE [19], and CLIFF [22], and extend to recent SMPL-X-based methods that improve temporal stability and controllability, including PromptHMR [43] and joint-feature-guided HMR [47], and CLIP-guided shape learning [26], with contemporary surveys summarizing the state of the art [30, 39]. Complementary efforts push for biomechanically faithful modeling, such as SKEL [18] and HSMR [45], as well as gait models like Generative GaitNet [33] and Bidirectional GaitNet [34] that condition on anatomical factors. Toolchains like OpenSim [9] and OpenCap [41] estimate kinematics and even kinetics from video, enabling musculoskeletal analyses outside the lab. Prior fitness/AQA systems [6, 11, 35, 37] still tend to output scores or templates, analyze frames rather than full cycles, and ignore morphology [25].

Structured Representations for Language Grounding. Structured interfaces for grounding language in perception span marker/slot prompts that tie text to localized evidence [46], layout- or schema-driven planners that impose explicit spatial structure before generation [12], and visual in-context prompting that conditions models on exemplar prompts across tasks [20]. Recent “visual chain-of-thought” work equips multimodal LMs with sketchpads to externalize intermediate spatial reasoning [14]. In human motion, motion-language models discretize continuous kinematics into tokens to align motion and text within a single generative interface [16]. Together, these directions show that explicit, compact structure can improve interpretability, controllability, and citation of evidence—key properties for corrective feedback on fine-grained physical behaviors.

3. Method

3.1. Overview

The proposed **BioCoach** framework converts streaming fitness videos into biomechanically-grounded coaching feedback by creating *explicit, interpretable intermediate representations* that bridge kinematic data and language generation. This design enables *transparent reasoning* about ex-

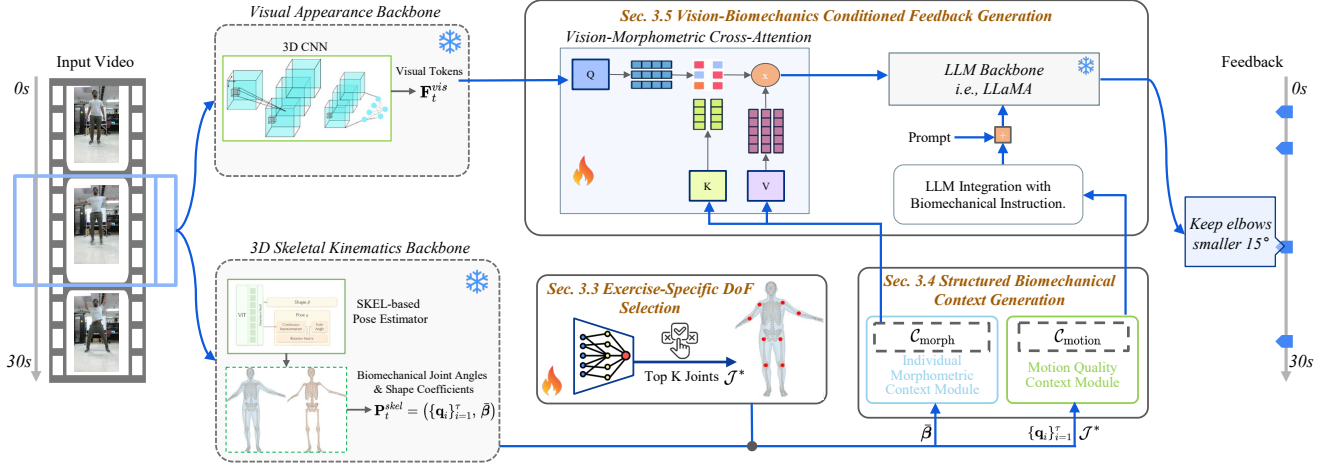


Figure 2. **BioCoach overview.** Streaming video is encoded by two backbones: a 3D CNN for visual tokens and a pose extractor for 3D skeletal kinematics. The pipeline has three components: (1) Exercise-Specific DoF Selection uses a lightweight attention head to select the top K biomechanically salient joints; (2) Structured Biomechanical Context builds two representations (individual morphometric context and motion quality context) capturing body measurements, cycles, ranges of motion, and constraint checks; (3) Vision-Biomechanics Conditioned Feedback fuses visual tokens with the morphometric context via cross-attention and prepends the motion-quality context as structured instruction to the LLM. This yields feedback grounded in explicit kinematic evidence rather than pattern matching alone.

ercise form, *personalization* to individual body geometry, and *feedback grounded in biomechanical principles* rather than learned patterns. As illustrated in Fig. 2, the framework extracts two complementary modalities from streaming video: visual appearance and 3D skeletal kinematics. These flow through a three-stage pipeline: (1) Exercise-Specific Degree-of-Freedom Selection Module (Sec. 3.3) identifies anatomically salient joints; (2) Structured Biomechanical Context Generation Module (Sec. 3.4) analyzes motion quality while accounting for individual body geometry; (3) Vision-Biomechanics Conditioned Feedback Generation (Sec. 3.5) generates feedback grounded in explicit biomechanical analysis.

3.2. Dual-Modality Feature Extraction

Our framework extracts two complementary modalities from streaming fitness videos. Both modalities originate from the video input $\mathbf{V} \in \mathbb{R}^{T \times H \times W \times 3}$ but capture different aspects of motion: visual appearance and 3D skeletal kinematics. To extract these modalities, we employ two specialized processing backbones, described below.

Visual Appearance Backbone. Following [32], we employ a pre-trained 3D CNN that captures fine-grained motion patterns essential for fitness coaching (see *Supp* for details). At each timestep t , the model processes a temporal window of video frames and extracts motion-aware features:

$$\mathbf{F}_t^{vis} = \mathcal{F}(\mathbf{V}_{[t-\tau:t]}), \quad (1)$$

where $\mathcal{F} : \mathbb{R}^{\tau \times H \times W \times 3} \rightarrow \mathbb{R}^{N_v \times d}$ is the 3D CNN feature

extractor, $\mathbf{V}_{[t-\tau:t]}$ denotes a sliding window of τ consecutive frames, and $\mathbf{F}_t^{vis} \in \mathbb{R}^{N_v \times d}$ contains N_v visual feature tokens with an embedding dimension of d . The architecture combines 2D and 3D convolutional layers: 2D layers capture spatial appearance cues within frames, while 3D layers encode temporal motion dynamics across the window. Crucially, all convolutions employ causal masking to ensure the model operates in a true streaming setting, generating predictions solely from past and current observations without access to future information.

3D Skeletal Kinematic Backbone. Complementary to visual appearance features, we extract explicit 3D skeletal kinematics to capture biomechanically-grounded body motion. We employ HSMR [45] built upon SKEL [18], a frame-level method that produces pose estimates in biomechanically-grounded representations. Following SKEL [18], we represent skeletal pose using 46-dimensional Euler-angle representations with joint-specific biomechanical constraints. To extract temporal kinematic information, we apply the pose extractor $\mathcal{P}$ to each frame in the temporal window $\mathbf{V}_{[t-\tau:t]}$:

$$\{\mathbf{q}_i\}_{i=1}^{\tau}, \{\beta_i\}_{i=1}^{\tau} = \mathcal{P}(\mathbf{V}_{[t-\tau:t]}), \quad (2)$$

where $\mathbf{q}_i \in \mathbb{R}^{46}$ represents biomechanical joint angles for frame i . To obtain a stable body shape representation across the temporal window, we aggregate per-frame shape coefficients via average pooling: $\beta = \frac{1}{\tau} \sum_{i=1}^{\tau} \beta_i$, expressed in SMPL-style parameterization. Thus, the kinematic modality output at time t is: $\mathbf{P}_t^{skel} = (\{\mathbf{q}_i\}_{i=1}^{\tau}, \beta)$, where

$\bar{\beta} \in \mathbb{R}^{10}$ encodes individual body shape. Unlike appearance features that conflate body geometry with motion quality, skeletal kinematics provide normalized, biomechanically-aware representations that separate body shape from movement patterns.

3.3. Exercise-Specific Degree-of-Freedom Selection

The biomechanical relevance of each joint depends on the exercise being performed. A squat requires lower-body joints (hips, knees, ankles), while a push-up demands upper-body joints (shoulders, elbows, wrists). This module identifies exercise-specific salient joints via visual context.

We leverage visual features $\mathbf{F}_t^{\text{vis}}$ to infer exercise-dependent joint relevance using a lightweight attention network $\mathcal{A}_\theta$:

$$\mathbf{s}^t = \mathcal{A}_\theta(\mathbf{F}_t^{\text{vis}}), \quad (3)$$

where $\mathcal{A}_\theta$ is an MLP with learnable parameters θ that outputs importance scores for each joint. Each element $\mathbf{s}_j^t \in [0, 1]$ represents the biomechanical relevance of joint j . We select the top- K joints by importance:

$$\mathcal{J}^* = \{j : \mathbf{s}_j^t \in \text{TopK}(\mathbf{s}^t, K)\}. \quad (4)$$

Skeleton joints correspond to multiple degrees of freedom in the 46-dimensional Euler angle representation $\mathbf{q}$ (e.g., the shoulder has 3 degrees of freedom (DoFs): flexion/extension, abduction/adduction, and internal/external rotation). When a joint is selected in $\mathcal{J}^*$, all its associated DoFs are automatically included in downstream analysis, ensuring coherent joint-level reasoning. The focused joint set $\mathcal{J}^*$ remains fixed for the entire exercise session, mimicking how human coaches maintain consistent attention to relevant body regions throughout a movement.

3.4. Structured Biomechanical Context Generation

This module generates subject-specific structured biomechanical contexts from kinematic data to enable personalized and biomechanically-grounded coaching. It consists of two submodules that, together, produce the individualized context required for the language module.

3.4.1. Individual Morphometric Context Module

Kinematic data must be contextualized within individual body geometry. This module extracts user-specific anthropometric measurements from the shape parameters $\bar{\beta}$ of the kinematic modality $\mathbf{P}_t^{\text{skel}}$. Raw shape parameters (i.e., SMPL [28] body shape coefficients $\bar{\beta}$) are abstract and difficult for language models to interpret. To address this, we leverage Virtual Measurements [7] to extract interpretable anthropometric measurements directly from the fitted SMPL mesh: mass, height, chest, waist, and hip circumference. Circumference measurements are derived via plane intersection, while length measurements use

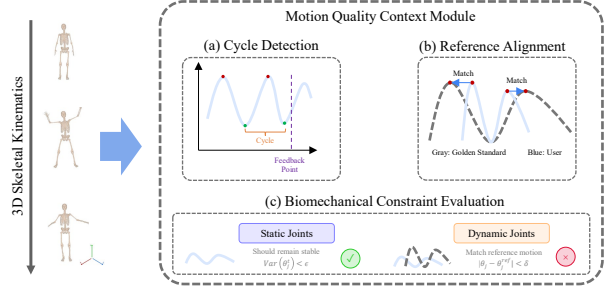


Figure 3. Motion-Quality Context module. Given the selected joint set and the 3D skeletal kinematics, the module (a) detects repetition cycles and anchors the feedback moment; (b) time-normalizes each cycle and aligns it to a curated reference trajectory; and (c) evaluates biomechanical constraints: stability for static joints and deviation to reference for dynamic joints. Gray curves denote the reference, blue curves denote the user.

Euclidean distance between anatomical landmarks [7]. This grounds shape in physical quantities that are semantically meaningful for exercise assessment. The extracted measurements form the morphometric context $\mathcal{C}_{\text{morph}}$ which is formatted as human-readable descriptors:

User body: height 1.78m, mass 73.22kg,
chest 1.00m, waist 0.83m, hips 0.98m.

By providing explicit, quantitative anchors grounded in individual body geometry, the morphometric context enables personalized and biomechanically-aware feedback. This representation is fused with visual features via cross-attention, allowing the language model to generate coaching feedback that accounts for individual differences.

3.4.2. Motion Quality Context Module

As illustrated in Fig. 3, this module evaluates whether the user’s motion satisfies biomechanical constraints for the given exercise, using the kinematic trajectory $\{\mathbf{q}_i\}_{i=1}^T$ and the selected joint set $\mathcal{J}^*$ from Sec.3.3, and produces structured descriptions of motion quality and form deviations.

Cycle detection. Motion cycles are identified by analyzing key joints in $\mathcal{J}^*$. At each frame i in the session, we have the full angle vector $\mathbf{q}_i \in \mathbb{R}^{46}$; we denote the primary DoF angle of joint $j \in \mathcal{J}^*$ at frame i as $q_{j,i}^*$, where $q_{j,i}^* = q_{d(j),i}$ and $d(j)$ maps joint j to its primary DoF index in $\mathbf{q}_i$ (e.g., $d(\text{knee}) = 7$ for knee flexion). For repetitive exercises (e.g., squats, push-ups), we select a representative joint j and extract its angle trajectory $\{q_{j,i}\}_{i=1}^N$ over the entire session. After gaussian smoothing, we identify cycle peaks and valleys using prominence-based peak detection, producing candidate cycle boundaries (i_s, i_e). Spurious detections are filtered by excluding cycles that are shorter than τ_{min} or

longer than $\tau_{\max}$. For exercise-specific variants: alternating movements (e.g., jumping jacks) use zero-crossing detection within the detected cycle. We compute the cycle-specific mean angle $\bar{q}_{j,\text{cycle}} = \frac{1}{i_e - i_s + 1} \sum_{i=i_s}^{i_e} q_{j,i}$ and detect zero-crossings of the normalized signal $q_{j,i} - \bar{q}_{j,\text{cycle}}$. Static holds (e.g., planks) identify low-variance regions where $\text{Var}(\{q_{j,i}\}_{i \in [i_s, i_e]}) < \epsilon$, indicating that the joint remains relatively stable within the detected cycle.

Reference Alignment. Each detected cycle (i_s, i_e) is temporally normalized by resampling to match a curated reference trajectory $\mathbf{R}$ via linear interpolation. For joint j , we interpolate sequences:

$$\tilde{q}_{j,k} = q_{j, \lfloor \phi(k) \rfloor} + (\phi(k) - \lfloor \phi(k) \rfloor)(q_{j, \lceil \phi(k) \rceil} - q_{j, \lfloor \phi(k) \rfloor}), \quad (5)$$

where $\phi(k)$ maps the reference index $k \in [1, N_{\text{ref}}]$ to the user's cycle timeline $i \in [i_s, i_e]$, and N_{ref} is the reference cycle length. Function ϕ rescales cycles to align with the reference length, ensuring that both trajectories map to a common temporal grid. The subscript j indicates the scalar angle trajectory for joint j , not the full 46-dimensional vector. We compute similarity scores by comparing the interpolated user cycle $\{\tilde{q}_{j,k}\}$ with the reference trajectory, which includes (1) cosine similarity on normalized angle profiles, (2) joint-specific Pearson correlations, (3) temporal derivative alignment measuring velocity consistency, and (4) amplitude comparison of the range of motion. These metrics yield a cycle quality score $s_{\text{cycle}} \in [0, 1]$.

Biomechanical Constraint Evaluation. For joint $j \in \mathcal{J}^*$, we evaluate exercise-specific biomechanical constraints based on the detected cycle (i_s, i_e) . Each joint is classified as *static* (should remain stable) or *dynamic* (should exhibit specific motion patterns), as determined by domain experts or LLM-assisted annotation (see **Supp** for details). For static joints, we measure stability via variance: $\delta_j^{\text{static}} = \text{Var}(\{q_{j,i}\}_{i \in [i_s, i_e]})$. For dynamic joints, we measure deviation from the reference trajectory at critical frames: $\delta_j^{\text{dynamic}} = |q_j^{\text{user}}(i_{\text{key}}) - q_j^{\text{ref}}(i_{\text{key}})|$, where i_{key} denotes critical frames (e.g., the deepest point of a squat). Violations are detected by checking against exercise-specific acceptable bounds $[l_j, u_j]$:

$$\text{violation}_j = \begin{cases} 1, & \text{if } \delta_j < l_j \text{ or } \delta_j > u_j \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

When detected ($\text{violation}_j = 1$), we generate descriptions quantifying deviations and corrections.

Structured Prompt Generation. At each frame, the module produces two outputs. Pose state captures selected joint configurations:

$$\mathbf{p}_{\text{state}}^{(i)} = \{\text{joint}_j \text{ angle } [q_{j,i}]^\circ \mid j \in \mathcal{J}^*\}, \quad (7)$$

For joints where violations are detected ($\text{violation}_j = 1$), we generate descriptions quantifying deviations:

$\mathbf{v}_{\text{violations}} = \{\text{"joint}_j \text{ deviation: } \delta_j \text{ (bound: } [l_j, u_j])\text{"} \mid \text{violation}_j = 1\}$. These are formatted as natural language:

Current pose: right knee 85°, left knee 88°, hip 75°.
Form issues: Right knee flexion insufficient (85° detected, 90° required)

Motion quality context concatenates pose state and violations:

$$\mathcal{C}_{\text{motion}} = [\mathbf{p}_{\text{state}}^{(i)}; \mathbf{v}_{\text{violations}}], \quad (8)$$

which enables precise feedback grounded in biomechanical analysis.

3.5. Vision-Biomechanics Conditioned Feedback Generation

This module synthesizes coaching feedback by conditioning a language model (i.e., LLaMA-2-7B [40]) on visual features $\mathbf{F}_t^{\text{vis}}$, morphometric context $\mathcal{C}_{\text{morph}}$, and motion quality context $\mathcal{C}_{\text{motion}}$.

Structured Multimodal Context Encoding. Both morphometric and motion contexts are encoded into token embeddings through the language model's embedding layer:

$$\mathbf{m}_t = \text{Embed}(\mathcal{C}_{\text{morph}}), \quad \mathbf{c}_t = \text{Embed}(\mathcal{C}_{\text{motion}}), \quad (9)$$

where $\mathbf{m}_t$ encodes body measurements and pose, and $\mathbf{c}_t$ encodes motion quality and constraint violations.

Vision-Morphometric Cross-Attention. Visual features $\mathbf{F}_t^{\text{vis}}$ are grounded in morphometric tokens $\mathbf{m}_t$ through cross-attention:

$$\mathbf{z}_t = \mathbf{F}_t^{\text{vis}} + \text{CrossAttn}(\mathbf{F}_t^{\text{vis}}, \mathbf{m}_t, \mathbf{m}_t), \quad (10)$$

where

$$\text{CrossAttn}(\mathbf{F}_t^{\text{vis}}, \mathbf{m}_t, \mathbf{m}_t) = \text{Softmax}\left(\frac{\mathbf{F}_t^{\text{vis}} W_Q (\mathbf{m}_t W_K)^\top}{\sqrt{d}}\right) \cdot (\mathbf{m}_t W_V), \quad (11)$$

with W_Q, W_K, W_V being learnable projection matrices. This produces fused visual-morphometric features $\mathbf{z}_t$ that align visual observations with individual body geometry.

LLM Integration with Biomechanical Instruction. Motion quality context $\mathcal{C}_{\text{motion}}$ encapsulates rich biomechanical analysis, including cycle-based motion quality, constraint violations, and quantified form deviations. This context is prepended as structured instruction to directly guide language generation with explicit biomechanical constraints:

$$\text{Prompt} = [\text{Embed}(\mathcal{C}_{\text{motion}}), \text{language_tokens}]. \quad (12)$$

The LLM then generates feedback conditioned on this explicit biomechanical analysis:

$$\text{Feedback}_t = \text{LLM}(\text{Prompt}, \{\mathbf{z}_t\}). \quad (13)$$

By injecting structured biomechanical constraints directly into the prompt rather than relying solely on learned patterns, this design ensures that feedback is grounded in explicit, exercise-specific biomechanical principles. (See *Supp* for mathematical details.)

3.6. Training Objective

We employ a parameter-efficient fine-tuning strategy that freezes both the 3D CNN visual backbone and the LLaMA-2-7B language model, updating only the cross-attention fusion layers and the DoF selection network $\mathcal{A}_\theta$. This design preserves pre-trained linguistic priors while adapting multi-modal fusion to align visual appearance, skeletal kinematics, and coaching feedback.

Exercise-Specific DoF Selection Training. The attention network $\mathcal{A}_\theta$ is trained with annotated exercise-specific joint relevance labels from domain experts and LLM-assisted annotation. We use binary cross-entropy loss to align predicted importance scores $\mathbf{s}^t$ with ground-truth salient joint sets $\mathcal{J}^{\text{gt}}$:

$$\mathcal{L}_{\text{DoF}} = - \sum_{j=1}^J [y_j \log(\mathbf{s}_j^t) + (1 - y_j) \log(1 - \mathbf{s}_j^t)], \quad (14)$$

where $y_j = 1$ for biomechanically-relevant joints and $y_j = 0$ otherwise.

Cross-Attention Fusion. Training mirrors the streaming inference regime using autoregressive cross-entropy loss with selective down-weighting to prevent excessive deferral of feedback:

$$\mathcal{L}_{\text{CE}} = - \sum_{t=1}^{N-1} w_{x_{t+1}} \log P(x_{t+1} | x_{\leq t}), \quad (15)$$

where $w_{x_{t+1}} = \alpha < 1$ for continuation tokens (`<next>` action tokens) and $w_{x_{t+1}} = 1$ for feedback content. This down-weighting encourages timely feedback generation rather than indefinite observation windows.

3.7. QEVD-bio-fit-coach Dataset

We augment QEVD-fit-coach [32] (149 training videos, 74 test videos, 23 exercises, $\sim 2,484$ timestamped feedback) to create **QEVD-bio-fit-coach**. The enhanced set adds biomechanically grounded feedback annotations. Using these signals, we systematically rewrite colloquial feedback into anatomically precise language: generic cues like “lower your body more” become quantified targets (“increase elbow flexion to 90° at the bottom”), and vague instructions become explicit alignments (“maintain a neutral spine, head to heels”). We also add brief biomechanical rationales (e.g., “increase hip/knee flexion to distribute load”). The temporal boundaries of all feedback instances are preserved exactly as in the original; only the *content* is rendered more

technical. This controlled augmentation isolates the effect of explicit biomechanical terminology and enables evaluation of whether kinematic grounding improves coaching quality.

3.8. Implementation Details

Following Stream-VLM [32], we adopt LLaMA-2-7B [40] as the language backbone and a pre-trained 3D CNN for visual features. Skeletal kinematics are represented by 46-dimensional Euler angles with a temporal window of 3 seconds ($\tau = 12$) of motion history to detect exercise cycles and evaluate biomechanical constraints. The DoF-selection network $\mathcal{A}_\theta$ is a 3-layer MLP (ReLU activations, sigmoid output) that produces per-joint importance scores $\mathbf{s}^t \in [0, 1]^J$; we select the top $K = 12$ salient joints per exercise. We train two variants: one on *QEVD-fit-coach* (original feedback) and one on *QEVD-bio-fit-coach* (biomechanics-grounded feedback). Both fine-tune only the cross-attention layers and $\mathcal{A}_\theta$ using AdamW (learning rate 2×10^{-5}), a batch size of 8, and action-token down-weighting $\alpha = 0.1$. DoF selection is supervised using a balanced binary cross-entropy loss. For biomechanical analysis, cycle durations are constrained to $[\tau_{\min}, \tau_{\max}] = [0.8, 5]$ s. Constraint thresholds are exercise-specific: static-joint variance $< 5^\circ$; dynamic-joint deviation tolerance $\pm 5^\circ$ to $\pm 10^\circ$ at key frames. Videos are resized to $H = 224, W = 160$ with standard augmentations. (See *Supp* for additional implementation details.)

4. Experiments

4.1. Experimental Setup

Dataset. We evaluate on QEVD-fit-coach [32], a streaming fitness-coaching benchmark with 74 test videos across 23 exercises and $\sim 2,484$ timestamped feedback instances. To better assess biomechanical understanding, we also report results on the *QEVD-bio-fit-coach*.

Baselines: We compare BioCoach against state-of-the-art vision–language models in two settings:

- Zero-shot models: InstructBLIP [8], Video-LLaVA [23], Video-ChatGPT [29], Video-LLaMA [49], LLaMA-VID [21], and LLaVA-NeXT [51]. These models are evaluated without any fitness-domain tuning and are periodically prompted to generate feedback from the video stream.
- Fine-tuned models: Methods trained on the QEVD-fit-coach benchmark [32]:
 - Socratic-LLaMA-2-7B [32]: A text-only baseline that feeds activity descriptors from a fine-tuned 3D CNN into LLaMA-2-7B to produce feedback.
 - Video-ChatGPT [29] and LLaMA-VID [21]: Turn-based VLMs fine-tuned on QEVD-FIT-COACH that

Table 1. Evaluation on the *QEVD-bio-fit-coach*, a newly created benchmark with fine-grained biomechanical ground-truth feedback annotations. Second line in each cell shows % improvement vs. Stream-VLM [32]. *LLM-Bio-Acc.* is our LLM-as-judge metric tailored to biomechanics, assessing the biomechanical correctness and specificity of generated feedback.

Method	METEOR $\uparrow$	ROUGE-L $\uparrow$	BERTScore $\uparrow$	LLM-Acc. $\uparrow$	LLM-Bio-Acc. $\uparrow$	T-F-Score $\uparrow$
Stream-VLM [32] (NeurIPS '24)	0.086	0.108	0.852	1.86	1.72	0.530
BioCoach	0.312 (+262.8%)	0.302 (+179.6%)	0.877 (+2.9%)	3.12 (+67.7%)	3.26 (+89.5%)	0.544 (+2.6%)

Table 2. Performance on the QEVD-fit-coach with original feedback annotations. Parenthesized values are % change vs. the best baseline (Stream-VLM [32]). $\dagger$ zero-shot without fine-tuning.

Method	METEOR $\uparrow$	ROUGE-L $\uparrow$	BERTScore $\uparrow$	LLM-Acc. $\uparrow$	T-F-Score $\uparrow$
<i>Zero-shot Models</i>					
InstructBLIP $\dagger$ [8]	0.047	0.040	0.839	1.56	-
Video-LLaVA $\dagger$ [23]	0.057	0.025	0.847	2.16	-
Video-ChatGPT $\dagger$ [29]	0.098	0.078	0.850	1.91	-
Video-LLaMA $\dagger$ [49]	0.101	0.077	0.859	1.29	-
LLaMA-VID $\dagger$ [21]	0.100	0.079	0.859	2.20	-
LLaVA-NeXT $\dagger$ [51]	0.104	0.078	0.858	2.27	-
<i>Fine-tuned Models</i>					
Socratic-LLaMA-2-7B [40]	0.094	0.071	0.860	2.17	0.50
Video-ChatGPT [29]	0.108	0.093	0.863	2.33	0.50
LLaMA-VID [21]	0.106	0.090	0.860	2.30	0.50
Stream-VLM [32] (NeurIPS '24)	<u>0.127</u>	<u>0.112</u>	<u>0.863</u>	<u>2.45</u>	0.56
BioCoach	0.129 (+1.6%)	0.122 (+8.9%)	0.864 (+0.1%)	2.56 (+4.5%)	0.544 (-2.9%)

use CLIP/ViT visual encoders and require prompting at fixed intervals.

- Stream-VLM [32]: The original streaming baseline on QEVD-FIT-COACH, combining a 3D CNN vision encoder with a LLaMA-2 language model and special `<next>`/`<feedback>` tokens for interactive, timely responses. This is the strongest, fine-tuned baseline tailored for fitness coaching.

Metrics. Following Stream-VLM [32], we report:

- METEOR [3]: lexical overlap with stemming, synonym, and paraphrase matching. Rewards correct technical terms and entities (*e.g.*, if the reference says “arm not moving,” predictions should mention the arm).
- ROUGE-L [24]: longest–common–subsequence overlap, capturing phrase ordering and multiword expression similarity.
- BERTScore [50]: semantic similarity via contextual embeddings and cosine similarity, robust to paraphrasing beyond exact word overlap.
- LLM-Accuracy [32]: LLaMA-3-70B-Instruct [10] acts as an automatic judge to holistically rate each feedback on a 1–5 scale (higher is better), considering context, correctness, and appropriateness; this correlates better with

human preferences than purely lexical metrics.

- Temporal F-Score (T-F-Score) [32]: time-alignment precision/recall F_1 indicating whether feedback is issued at the correct moments.

In addition, to evaluate biomechanical fidelity on QEVD-bio-fit-coach, we introduce LLM-Bio-Accuracy, an LLM-as-judge metric specialized for biomechanics. Using the same LLaMA-3-70B-Instruct with a domain-specific prompt, it scores (1–5) factual biomechanical correctness and relevance against the reference context (implicitly reflecting per-frame kinematics, phase labels, and constraint annotations), emphasizing precise anatomy (joint names/sides), angles, range of motion, and phase-aware reasoning (see *Supp* for more details).

4.2. Results on QEVD-bio-fit-coach

To assess biomechanics-grounded feedback on streaming video, we evaluate BioCoach on the new QEVD-bio-fit-coach benchmark and compare it to the strongest streaming baseline, Stream-VLM [32], which has been fine-tuned on the same annotations for fairness (Tab. 1). This benchmark stresses anatomy/kinematics-centric language, quantitative ranges of motion, and phase-aware cues—precisely what safe, actionable coaching requires.

Table 3. Ablation study on *QEVD-Bio-Fit-Coach*. Each row removes or modifies one component while keeping others fixed.

Model Variant	METEOR $\uparrow$	ROUGE-L $\uparrow$	BERTScore $\uparrow$	LLM-Acc. $\uparrow$	LLM-Bio-Acc. $\uparrow$	T-F-Score $\uparrow$
Full Model ($\tau = 3$ s)	0.312	0.302	0.877	3.12	3.26	0.544
w/o Exercise-Specific DoF Selection	0.305	0.296	0.873	3.06	3.14	0.543
w/o Motion Quality Context Module	0.133	0.127	0.864	2.63	2.04	0.544
w/o Individual Morph Context Module	0.284	0.273	0.871	2.91	3.07	0.535
Window size $\tau = 2$ s	0.311	0.303	0.876	3.10	3.22	0.416

BioCoach markedly outperforms Stream-VLM across all metrics, with the largest relative gains in lexical measures (METEOR +262.8% and ROUGE-L +179.6%), reflecting improved alignment with expert references. Judge-based evaluations also rise sharply: *LLM-Bio-Acc.* shows the strongest improvement (+89.5%) among judge metrics, indicating biomechanics-grounded, anatomy- and phase-accurate feedback. Temporal behavior remains strong (T-F-Score), showing that richer biomechanical grounding does not come at the expense of timely triggering. These gains stem from exercise-focused joint selection, personalized morphometrics, and cycle-aligned analysis that turn raw kinematics into verifiable cues, shifting the system from pattern matching to principled, auditable reasoning.

For a fair comparison of Stream-VLM [32], we re-train BioCoach on QEVD-fit-coach (with the original annotations), *i.e.*, without any of our biomechanical labels, and evaluate it under the official protocol. This experiment serves two purposes: (i) it isolates architectural gains from annotation gains, showing that our biomechanics-grounded design helps even when trained with non-biomechanical labels; and (ii) it demonstrates *backward compatibility*, indicating BioCoach can be deployed on datasets that lack fine-grained biomechanical supervision. Under this setting (Tab. 2), BioCoach still surpasses the strongest fine-tuned baseline in text/semantic quality and judged correctness (METEOR 0.129, ROUGE-L 0.122, BERTScore 0.864, LLM-Acc. 2.56), while remaining near-parity in timing (T-F-Score 0.544 vs. 0.56). Together with the large improvements in QEVD-bio-fit-coach, these results show that (a) BioCoach improves quality under legacy annotations and (b) adding biomechanical supervision further unlocks its full potential for precise, anatomy-specific feedback.

4.3. Ablation Study

Ablations on QEVD-bio-fit-coach (Tab. 3) reveal a clear component hierarchy. Removing Motion Quality Context causes $\sim 57\%$ METEOR drop and LLM-Bio-Acc. decline ($3.26 \rightarrow 2.04$), showing it is the core biomechanical driver. Removing Morphometric Context causes moderate degradation (-9% METEOR, -6% LLM-Bio-Acc), indicating that personalization improves specificity. DoF Selection

provides modest gains (-3.7% LLM-Bio-Acc without it), filtering irrelevant kinematics. Reducing the temporal window to 2 s maintains text quality but drops the T-F-Score $\sim 24\%$, confirming that 3 s offers better stability for cycle detection. The three modules divide labor: motion quality supplies *what to say*, morphometrics supply *for whom*, and DoF selection supplies *where to look*—together enabling interpretable reasoning without sacrificing responsiveness.

4.4. Limitations and Future Work

Our approach depends on the quality of 3D skeletons and shapes, as occlusions, loose clothing, and extreme viewpoints can distort kinematics. Additionally, it relies on curated reference trajectories that may miss exercise variants and adaptive forms. Future work will explore multi-sensor fusion and broader reference coverage to mitigate these issues. Beyond kinematics, we plan to extend BioCoach toward kinetic reasoning, estimating joint reaction forces and muscle activation patterns from video to detect compensatory movement strategies that are invisible to angle-only analysis. Integrating musculoskeletal simulation could further enable load-aware coaching that flags injury-risk forces, not just postural deviations.

5. Conclusion

We present **BioCoach**, a biomechanics-grounded vision-language framework for streaming fitness coaching that exposes 3D kinematics, morphometrics, and constraint analyses to the language model. With an exercise-specific DoF selector, a morphometric and motion-quality context, and a vision-biomechanics conditioned feedback module, BioCoach delivers interpretable coaching rather than appearance-level heuristics. BioCoach improves lexical, semantic, and judge metrics while maintaining competitive timing, demonstrating that explicit kinematics and structured biomechanical context are key to accurate, phase-aware, personalized coaching. This formulation opens a path to coaching systems whose behavior can be inspected, trusted, and aligned with expert musculoskeletal reasoning.

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